Automation of Welding Using Cobots

1 Introduction

Welding is a widely used manufacturing process that joins metal components by melting a filler wire using a high-energy heat source, typically an electric arc. The molten material solidifies to form a permanent joint. Arc-based welding methods, particularly Metal Inert Gas (MIG) welding, are common due to their high deposition rates, relative ease of automation, and cost-effectiveness [1,2] Although welding is one of the most common manufacturing activities in the heavy industry and transportation sectors, much of it is currently performed manually by human operators. In spite of its importance, there is little progress in terms of automation of welding, especially in freeform welding of complex parts and assemblies with multiple contours requiring complex weld paths and long-line welds where human operators are subject to considerable fatigue and other health-related issues. Automation of welding using cobots is seen as a potential alternative to manual free-form welding of complex assemblies, where the human operator can work in close proximity to the robot for welding setup, part registration, and, if necessary, the ability to perform finishing welds. Cobotic systems offer greater precision, repeatability, and throughput than manual welding. The throughput improvements and flexibility of quickly repurposing cobots for other manufacturing tasks offer significant benefits to small- and medium-scale manufacturers. By automating the execution of the weld path, these systems reduce human error, improve safety, and enable consistent weld quality.

1.1 Challenges in Robotic Welding

Despite the advantages of robotic welding, several technical challenges must be addressed to ensure consistent and reliable performance. A fundamental requirement in robotic welding is accurate part registration. Unlike a human welder, who can visually identify the position and orientation of a workpiece, a robot must be explicitly informed of the pose of the part (position and orientation) relative to its base frame. Any error in this transformation directly affects the accuracy of the planned weld path and may lead to misaligned or missing welds.

Another key challenge is collision avoidance. During welding, the robot must navigate the part without colliding with the workpiece, fixtures, or itself. This requires careful planning of approach, weld and retreat paths that respect the physical constraints of the environment and the robot's kinematic limits. Trajectories must be constructed to avoid both static and dynamic obstacles while maintaining the required torch orientation.

Maintaining a proper tool orientation is also critical for ensuring weld quality. For example, fillet welds typically require the welding gun to be oriented at a 45-degree angle to the joint. The robot must consistently hold this angle throughout each weld segment, even as it moves along curved or multi-directional paths. Inconsistent orientation can lead to poor fusion or asymmetric weld beads.

In addition, the robot must maintain a constant travel speed along the weld path. Any variation in the weld tip speed from the desired can affect the amount of heat input, which in turn impacts the penetration depth, bead geometry, and defect formation. Achieving consistent speed requires the development of a robot trajectory execution with proper velocity profiling and synchronization across joints.

Finally, to make robotic welding viable in practice, especially for use by small to medium-scale manufacturers, the entire workflow must be easy to setup and use. The user should not be required to perform extensive programming or calibration. Instead, the system should provide a unified

interface, such as a graphical user interface (GUI), where weld paths can be selected, visualized, and executed with minimal effort. A streamlined interface ensures repeatability, reduces setup time, and lowers the barrier to adoption for non-expert users.

1.2 Literature review

1.2.1 Welding Overview

Welding methods are classified as arc, spot, gas, friction, or beam welding, with arc welding being the most dominant in industrial applications. Arc welding is widely adopted due to its rapid heating capabilities and its ability to reduce the heat affected zone (HAZ), which is inversely proportional to input power. This process applies electrical current to generate heat, melting a filler material to join metal components. Spot welding, traditionally popular in automotive manufacturing, uses current passed between two electrodes to fuse parts: an average car may contain about 5000 spot welds [3, 4].

In recent years, robotic arc welding has become more common. The most widely used forms are metal-inert gas (MIG) and tungsten-inert gas (TIG) welding. MIG welding employs a consumable electrode, making it more compatible with robotic automation, as it allows for a self-contained welding end effector. TIG welding, on the other hand, may require additional tools or even human-robot interaction. Although laser welding offers extremely precise results with a minimal HAZ, its high cost limits its feasibility to large-scale operations. In contrast, MIG welding is both cost-effective and straightforward, making it the preferred method for smaller manufacturers.

Several parameters influence the quality of the weld and vary depending on the process. An issue specific to MIG welding is the contact tip burnback, which occurs when the wire melts back to the contact tip due to arcing within the nozzle [5]. This can be mitigated by increasing the wire feed rate; however, higher wire tension can increase the stress on the wire feed motor, while too little leads to feeding failures [6]. Higher current levels require the contact tip to be recessed further into the nozzle, increasing wire adhesion, reducing heat at the tip, and improving gas flow [7].

Other common issues include excessive spatter, porosity, and insufficient or excessive penetration. The spatter of material from the weld pool (molten droplets leaving the weld pool) is affected by voltage, current, arc length, weld angle and gas shielding [8]. Porosity results from contamination by atmospheric gases (nitrogen, oxygen, hydrogen), forming cavities that weaken the weld; proper gas flow rate and mixture are essential to avoid it. Inadequate penetration of the weld produces weak joints, while excessive penetration can cause distortion and structural instability [9].

Weld quality optimization involves fine-tuning all these parameters, usually beginning with recommended specifications and iteratively adjusted based on visual inspection and material properties. Additional factors such as the travel speed of the weld tip, the joint geometry, surface preparation, and ambient temperature must also be considered.

1.2.2 Path Planning and Current Solutions

Path planning refers to the process of generating collision-free trajectories that enable the robot to move from an initial configuration to a desired target while satisfying kinematic and workspace constraints. A common strategy involves sampling a set of spatial points, filtering them based on geometric constraints, and then connecting valid pairs to form feasible paths. Triangulation techniques, such as Delaunay triangulation, are often used to define a set of safe interior regions for sampling. These sampled points can then be used to construct a graph, where each edge represents a valid transition. Graph search algorithms such as Dijkstra's and A* are then applied to find an optimal sequence of waypoints along this graph [10], [11].

While these techniques are effective in generating feasible Cartesian paths, robotic manipulators require corresponding joint-space trajectories. Therefore, each Cartesian pose must be mapped to a valid joint configuration via inverse kinematics (IK). For a 6-DOF robot, this mapping is typically non-unique, and selecting appropriate joint configurations that maintain continuity is critical. Furthermore, improper handling of angle wrapping (e.g., transitions between 0 and 2π radians) can result in high-jerk motions and poor weld consistency. These issues are typically resolved through iterative solvers such as the Newton-Raphson method, followed by angle unwrapping and post-processing to ensure smooth motion across all joints [12].

Several industrial solutions have been developed to automate welding using collaborative robots. Hirebotics, for example, introduced a UR10e-based welding system designed for ease of use in small-to-medium-scale manufacturing [4]. In a case study with Vortex Companies, they demonstrated a reduction in weld time from approximately 60 minutes to just 12 minutes for a single component. Their system includes a teach-free interface using a button-mounted torch, allowing welds to be stored and replayed without manual path programming. However, the system does not support dynamic path generation based on CAD input and requires manual teaching for each new part.

A more advanced solution is provided by Path Robotics, which developed an AI-enabled welding platform with real-time path planning and adaptive control [13]. Their system includes multiple collaborative arms: one pair of UR arms used for part handling and a Yaskawa arm for executing the weld using a Miller Auto Continuum 500 welder. The end-effector integrates a laser vision sensor capable of scanning the part geometry and dynamically adjusting weld parameters based on detected variations. Additionally, the use of a rotating fixture table enables multi-axis access for curved weld seams and non-planar geometries. The platform is commercially deployed as a subscription service and eliminates the need for manual path programming or calibration.

1.3 Collaborative Robots in Welding

Cobots bridge the gap between traditional and industrial robots-which often lack safety features and require complex setup—and the need for fast, adaptable automation. Cobots provide safe, cost-effective, and flexible automation for manufacturing environment, which allows them to share the same workspace with humans. Their intuitive programming interfaces also allow operators to quickly modify or update tasks as needed.

This report describes a systematic development of the framework and the technical tools needed to facilitate cobotic welding. It presents a step-by-step framework for robotic welding, including part registration, weld path generation, trajectory planning, process control, motion control, and GUI development. Figure 1 illustrates the full system workflow, from part import and registration to simulation and robotic weld execution.

2 System Setup

The experimental robotic welding setup consists of a UR10e collaborative robot and a remotely controlled MIG welding system based on the Miller Invision 352 MPa welder. The Invision 352 manages the welding power supply and interfaces with the central computer through the Miller Insight Core Module using a 14-pin connection. This configuration is integrated with a Miller 74 Series MPa Drive Assembly and an S-74 MPa Control Box, which deliver the required filler wire and shielding gas to a Tregaskiss BA1 Cobot Air-Cooled MIG gun mounted on the robot.

The system uses a 0.035 inch diameter mild steel wire, which serves both as an electrode and as a filler material. It is capable of welding materials with thicknesses ranging from 1.2 to 12.7

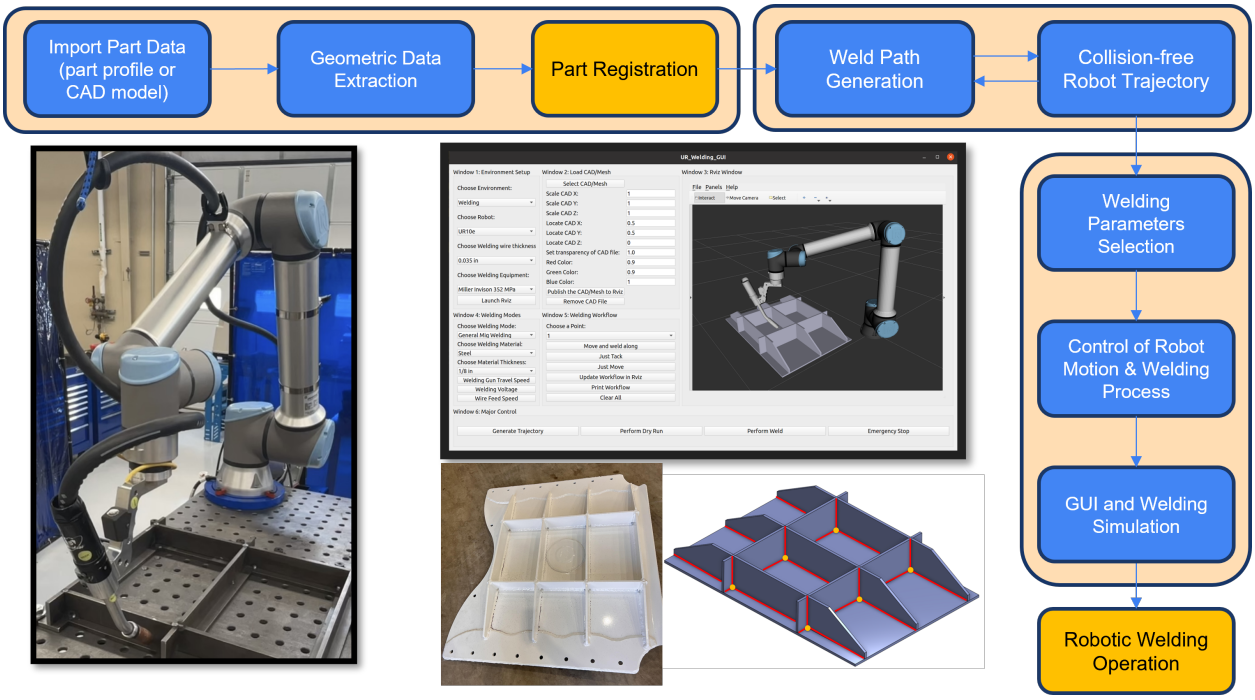

Figure 1: Overview of the cobot-based robotic welding workflow. The system includes part data import, registration, weld path generation, trajectory planning, parameter control, GUI interface, and final robotic weld execution.

mm. Two welding modes are supported: spray transfer and pulsed MIG. The spray transfer mode operates at a constant current, creating a continuous stream of molten metal droplets. In contrast, pulsed MIG periodically varies the current to deposit individual droplets, offering better control over heat input.

Pulsed MIG was chosen as the default welding mode because of its improved bead quality and reduced heat-affected zone (HAZ), which is especially beneficial for thin materials and thermally sensitive joints. The welder also supports fine-grained control over key parameters, including wire feed rate, arc voltage, and programmable lead-in/lead-out ramps. These ramps allow for smooth transitions in current, voltage, and feed rate at the beginning and end of a weld, helping to reduce defects caused by abrupt thermal changes.

A simplified electrical communication diagram for the Arduino-based controller used to trigger and modulate welder commands is shown in Figure 2.

3 Part Registration

This section describes the procedure used to register the part in the robot's coordinate frame, which is a prerequisite for accurate weld part execution. The registration process establishes a transformation between the part model (CAD-based) and the physical location of the part in the robot's workspace.

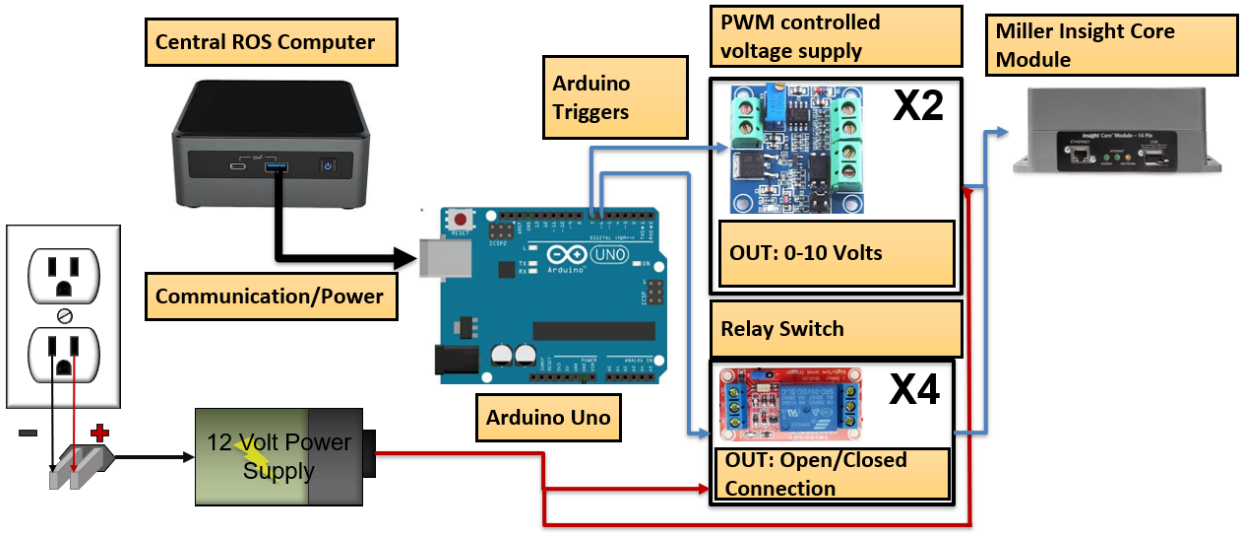

Figure 2: Simplified communication diagram for the Arduino-based controller used to interface with the Miller welding system.

3.1 Prototype Test Part

The target application involves robotic welding of a production component used by Exosent Engineering. This part consists of a complex lattice of intersecting vertical and horizontal plates, designed for structural reinforcement in pressure vessel housings. A CAD view of the actual Exosent part is shown in figure 3 (left).

To enable early testing and avoid excessive material waste, a simplified prototype was fabricated. The prototype maintains the essential features required to validate part registration, weld path generation, and motion control. It consists of eight flat steel plates arranged to form a rectangular enclosure. Specifically, six plates measure $9.5 \times 2 \times 3/8$ inches, and two measure $8 \times 2 \times 3/8$ inches. This structure is shown in figure 3 (right).

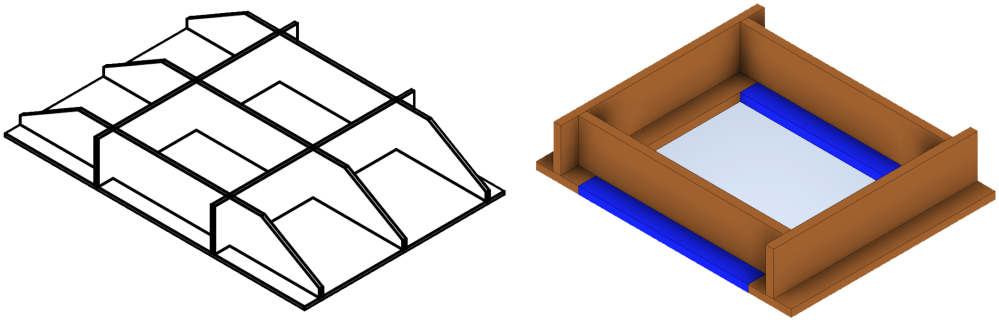

Figure 3: Left – Exosent production part; Right – simplified prototype used for testing

3.2 Registration Setup

The cobot is equipped with a Tregaskiss BA1 welding gun, which serves as the main end effector throughout the process. No additional tooling or sensing hardware is required for registration. This approach leverages the same equipment used for welding, significantly reducing the complexity and cost of the system.

The collaborative feature of the UR10e allows the operator to manually guide the robot using free-drive mode. This feature enables direct physical manipulation of the robot arm without having to interface through software or teach pendant commands. Using this capability, the user can move the tool center point (TCP) freely and precisely to desired positions on the physical part without requiring additional external measurement systems or calibration devices.

To register the part, three noncollinear points are first selected in the CAD model. The operator then manually moves the robot to align the welding tip with the corresponding physical points on the part. In each position, the internal encoders of the robot are used to record the 3D location of the TCP relative to the base of the robot. These data form the input to the transformation computation step. Choosing three non-collinear points ensures that the resulting transformation matrix captures both rotational and translational differences between the model and physical frames. The three-point correspondence provides enough geometric information to compute a full rigid-body transformation. An example of three selected points is shown in Figure 4.

3.3 Transformation Computation using Kabsch Algorithm

The Kabsch algorithm is used to compute the optimal rigid-body transformation that aligns the three registered points in the robot frame to the corresponding CAD model points. This method minimizes the root mean square deviation (RMSD) between the matched points under rotation and translation. The output is a 4×4 homogeneous transformation matrix $T_{part \to robot}$, which can be used to transform all points from the CAD model to the base frame of the robot.

Once this transformation is established, every weld point or trajectory defined in the local frame of the part can be transformed to the robot workspace. This allows subsequent trajectory planning, inverse kinematics, and execution to proceed using coordinates that are consistent with the physical setup.

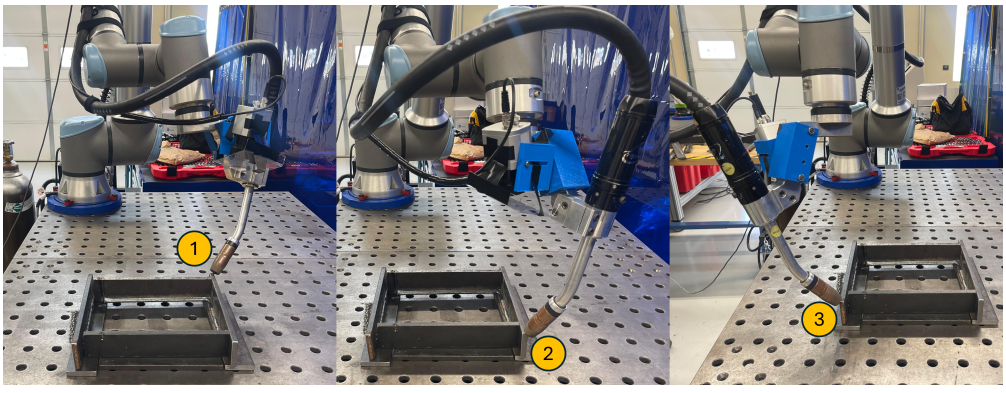

Figure 4: Three noncollinear points picked for part registration

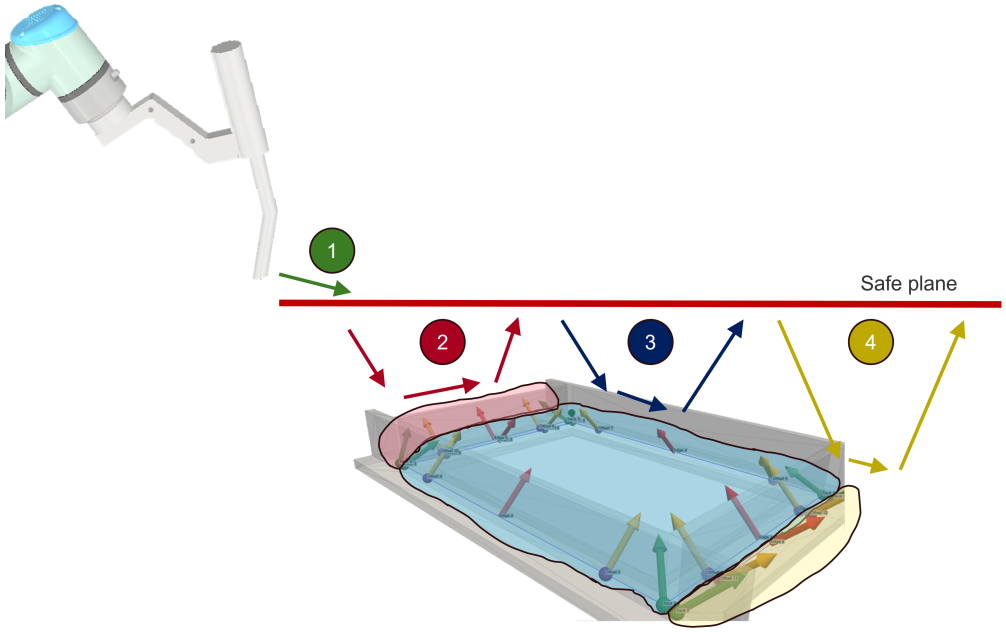

Figure 5: Weld trajectory execution with section-based segmentation. The weld path is divided into connected sections (2, 3, 4), each executed after the robot descends from the safe plane (1). The robot follows a approach → weld → retreat → reorient motion for each section.

4 Weld Path and Robot Trajectory Generation

This section describes the automatic generation of weld paths and corresponding robot trajectories using a CAD model of the part. The user only needs to select the regions of the part that require welding, and all subsequent computations are handled by the system.

4.1 Weld Path Generation

The first step in the process is user interaction with the CAD model to specify which weld seams are desired. The user selects discrete points along the edges of the part that should be welded. This is the only manual input required to define the weld path. Once the points are selected, the program identifies the edges in the mesh that connect two or more selected points. If a continuous edge exists between them, that entire segment is designated as a weld path.

After defining the weld segments, the system computes the desired orientation of the end effector at each weld point. These orientations are derived on the basis of surface normals and relative direction of the weld lines. For linear segments, the orientation vector is aligned along the weld path direction. The welding gun is tilted at an angle of 45 degrees relative to the surface of the part to ensure proper fusion and consistent quality of the beads.

The selected points are then grouped into contiguous sections. Each section consists of points that are directly connected by weldable edges. This segmentation enables the execution of continuous welds within each section, minimizing unnecessary stopping and restarting of the welding process. Figure 5 shows the segmentation of the weld path into multiple connected sections, along with the safe plane defined above the workpiece.

4.2 Robot Trajectory Planning

To perform the welding process without collisions, a safe travel strategy is implemented. A safe plane is defined parallel to the surface of the part at a fixed height above the highest point of the workpiece. This plane serves as a collision-free region where the robot can reposition and reorient itself freely between weld sections.

The trajectory of each weld section follows a structured four-stage pattern. First, during the *approach* phase, the robot descends from the safe plane to the start point of the weld while maintaining the required end-effector orientation. Next, in the *weld* phase, a Cartesian path is executed along the weld segment with constant tip velocity and orientation control to ensure consistent bead quality. Upon completion of the weld, the robot enters the phase *retreat*, where it retracts vertically to return to the safe plane. Finally, in the *reorient* phase, the robot adjusts its orientation in the safe plane to align itself for the next weld section or to relieve any accumulated cable tension before beginning the next cycle.

This approach ensures smooth transitions, consistent weld quality, and avoidance of singularities and collisions throughout the weld tour.

5 In-situ Weld Pool Measurement

To enable real-time monitoring and control of the welding process, an Optris PI 05M infrared camera is mounted directly on the welding gun. The camera is integrated with the weld gun housing, ensuring that it remains fixed relative to the torch tip and does not protrude beyond the tool's outer profile. This placement guarantees that the camera consistently views the weld pool during operation while also preventing any risk of collision with the workpiece, regardless of changes in the robot's orientation or joint configuration. Figure 6 shows the physical and CAD views of the camera integrated with the welding torch.

5.1 Sensor Specifications and Capabilities

The Optris PI 05M is a short-wavelength infrared thermal camera optimized for high-temperature applications such as metal welding. It operates in the spectral range of 0.50 to 0.54 µm and can measure temperatures between 900 °C and 2450 °C. The camera features a high dynamic range CMOS detector with a resolution of up to 764 × 480 pixels, enabling precise, non-contact measurement of thermal characteristics in high-speed processes.

These features make the sensor suitable for capturing both geometric and thermal features of the molten weld pool during arc welding.

5.2 Weld Pool Height Measurement

The weld height is extracted from thermal image frames using edge detection. Specifically, the top and bottom boundaries of the glowing weld region are identified. The vertical pixel distance between these edges is then converted to a physical height using a known pixel-to-millimeter ratio. This conversion accounts for the camera mounting angle and the standoff distance from the workpiece. Importantly, edge detection is performed slightly downstream of the arc to avoid saturation from the bright core and to ensure reliable boundary detection. An example of the edges of the detected weld pool is shown in Figure 7, where the upper and lower boundaries are highlighted for height calculation.

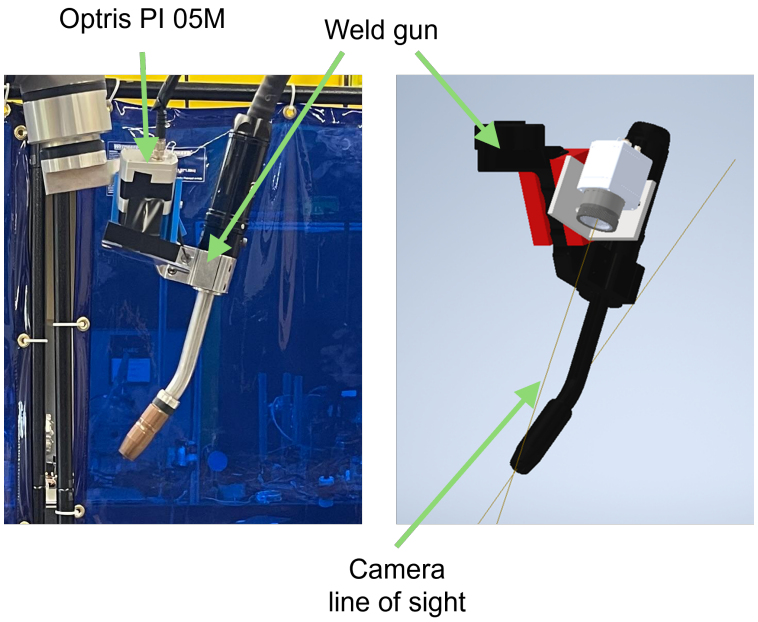

Figure 6: Left: Physical setup of the Optris PI 05M camera mounted on the welding gun. Right: CAD rendering showing camera alignment and compact integration within the torch assembly. The mounting ensures a fixed field of view while avoiding collisions.

5.3 Weld Pool Temperature Measurement

Temperature measurement is performed directly from the calibrated thermal images produced by the camera. The sensor's narrow spectral range in the near-infrared band allows it to avoid ambient infrared interference and focus on the high-emissivity region of molten metal. The temperature data are used to assess the thermal state of the weld pool and track its consistency throughout the process.

5.4 Importance of Height and Temperature Monitoring

Maintaining consistent weld pool height is essential for achieving uniform bead geometry and structural integrity. Deviations in height can result in underfilling, excessive reinforcement, or incomplete fusion. Likewise, precise regulation of the weld pool temperature is critical for controlling penetration depth and ensuring microstructural quality. Excessive temperatures can cause undesirable effects such as porosity, grain coarsening, and thermal distortion or warping of the base material.

These two parameters, height and temperature, serve as key feedback signals in the control loop. The system is designed to eventually regulate the weld tip travel speed and arc voltage in real time based on this feedback, enabling closed-loop control of the process.

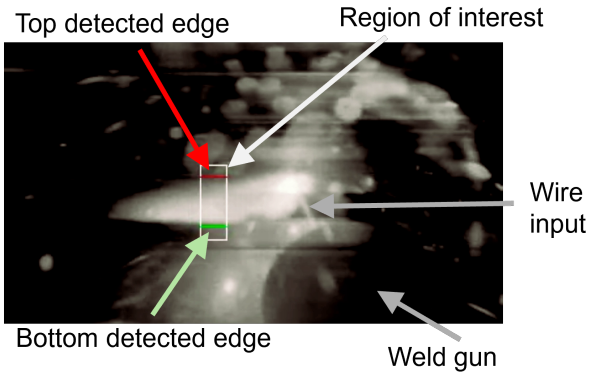

Figure 7: Thermal image frame showing detected weld pool boundaries. The top (red) and bottom (green) edges are used to estimate weld pool height.

6 Robot Motion Control

6.1 Objective

The primary objective of the robot motion control module is to ensure an accurate and consistent execution of the welding trajectory by the robotic manipulator. Specifically, the end effector must follow a predefined Cartesian path composed of spatial waypoints with precise orientation and position. Maintaining a constant linear velocity along the weld path is critical, as it ensures uniform heat input and consistent weld seam formation.

Additionally, the motion of the robot must be smooth and continuous, avoiding abrupt changes in joint angles that could cause mechanical limits, poor weld quality. To preserve this continuity, the trajectory planning must eliminate angular discontinuities across joints, particularly in multi-directional welds.

The system must also automatically handle joint limit violations, especially during the execution of complex paths where certain joints approach mechanical constraints. If such conditions are detected, the trajectory should be dynamically adjusted to remain within safe operating limits without compromising weld quality. This integrated pipeline is essential for maintaining high welding quality and reliable execution in varied workpiece geometries.

6.2 Trajectory Generation

The input to the motion controller is a sequence of Cartesain poses, each consisting of a 3D position and a 3D orientation vector. Each pose is converted to a rigid transformation for the desired frame construction. These are then mapped to the joint-space configuration trajectory using optimal inverse kinematics(Newton- Raphson IK) integrated with ROS.

6.3 Trajectory Execution

The trajectory execution process ensures that the robotic manipulator performs welding tasks with consistent motion, safe joint behavior, and reliable path tracking. The motion is structured into three stages: an initial approach to the weld seam, a controlled welding motion along the seam, and a final retreat. To maintain a uniform welding height, the Cartesian poses along the weld are constrained to a common plane, typically by averaging their vertical coordinates. Spatial resolution

is improved by inserting interpolated poses between the original waypoints. These interpolations, including both position and orientation, promote smoother transitions and more stable tool motion.

The Cartesian path is then mapped into joint space using Newton-Raphson inverse kinematics, and the timing of each trajectory point is determined based on the spatial distance between poses and a specified linear velocity. This ensures that the tool moves with approximately constant speed, which is essential to control heat input and maintain weld consistency. The system also includes a mechanism to detect and handle joint limit violations, especially the wrist joint, which can accumulate excessive rotation over long or looped welds. If such a condition is detected, the motion is temporarily halted, the joint configuration is reset to a safe value, and the remaining motion is resumed. This approach prevents the robot from reaching mechanical limits and preserves consistacy in the weld pool.

In general, the execution process emphasizes smooth and reliable motion, uniform tool velocity, and adherence to mechanical limits, enabling precise and repeatable robotic welding.

6.4 Construction of Frames

To define the orientation of the robot end effector at each weld point, a right-handed orthonormal frame is constructed using the weld direction vector. Each weld pose is specified by a position vector $\mathbf{p} \in \mathbb{R}^3$ and an orientation direction vector $\mathbf{z} \in \mathbb{R}^3$, where $\mathbf{z}$ represents the desired tool Z-axis direction (that is, the weld direction). This direction vector is first normalized:

$$\hat{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|}$$

To complete the frame, an orthogonal vector $\hat{\mathbf{x}}$ is computed by taking the cross product of a world-up reference vector (typically $\mathbf{u} = [0,\, 0,\, 1]^\top$) with the normalized Z-axis:

$$\mathbf{x} = \mathbf{u} \times \hat{\mathbf{z}}$$

In the case where $\hat{\mathbf{z}}$ is nearly parallel to $\mathbf{u}$ (that is, $\|\mathbf{x}\| < \varepsilon$ for a small threshold ε), the cross product becomes numerically unstable and a fallback direction is used.

$$\mathbf{x} = \begin{cases} \frac{\mathbf{x}}{\|\mathbf{x}\|}, & \text{if } \|\mathbf{x}\| \geq \varepsilon \\ [1,\, 0,\, 0]^\top, & \text{otherwise} \end{cases}$$

To maintain smooth orientation transitions across consecutive poses, a frame continuity check is performed. If there is a previous X-axis direction $\mathbf{x}_{\text{prev}}$ and the dot product $\hat{\mathbf{x}} \cdot \mathbf{x}_{\text{prev}} < 0$, then the new X-axis is flipped to align directionally:

$$\hat{\mathbf{x}} \leftarrow -\hat{\mathbf{x}} \quad \text{if } \hat{\mathbf{x}}^\top \mathbf{x}_{\text{prev}} < 0$$

The Y-axis of the frame is then determined by the cross product of the Z and X axes:

$$\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$$

These three unit vectors form the rotation matrix $\mathbf{R}$:

$$\mathbf{R} = \begin{bmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \end{bmatrix}$$

This rotation matrix is then converted to a unit quaternion $\mathbf{q} \in \mathbb{R}^4$. The final pose $\mathcal{T}$ is then defined by the translation vector $\mathbf{p}$ and quaternion $\mathbf{q}$:

$$\mathcal{T} = \{\mathbf{p},\ \mathbf{q}(\mathbf{R})\}$$

This pose is returned along with the current $\hat{\mathbf{x}}$ axis to be used for continuity in frame construction for subsequent poses.

6.5 Optimal Planner

The optimal planner is responsible for generating a smooth, time-parameterized joint-space trajectory that satisfies a given sequence of Cartesian end-effector poses while ensuring constant linear tool-tip velocity and joint motion continuity. The planner integrates two key components:(1) an inverse kinematics solver based on the Newton–Raphson method to compute joint configurations corresponding to each pose; (2) a sampling-based motion planning algorithm to generate feasible collision-free joint paths.

6.5.1 Inverse Kinematics using Newton–Raphson

A feasible joint configuration is found for the desired end effector pose $\mathcal{T}_d \in \mathrm{SE}(3)$ using an iterative inverse kinematics method. Let $f : \mathbb{R}^n \to \mathrm{SE}(3)$ be the forward kinematics map. The goal is to find $\boldsymbol{\theta}_d \in \mathbb{R}^n$ such that:

$$f(\boldsymbol{\theta}_d) = \mathcal{T}_d$$

Define the error function as:

$$g(\boldsymbol{\theta}) = \mathbf{x}_d - f(\boldsymbol{\theta})$$

where $\mathbf{x}_d$ is the position and orientation vector (e.g., expressed as a 6D twist).

At each iteration k, a linear approximation of g yields the following.

$$g(\boldsymbol{\theta}^{(k+1)}) \approx g(\boldsymbol{\theta}^{(k)}) + J(\boldsymbol{\theta}^{(k)})\Delta\boldsymbol{\theta}^{(k)}$$

where $J(\boldsymbol{\theta}^{(k)}) \in \mathbb{R}^{6\times n}$ is the geometric Jacobian.

Solving for the correction:

$$\Delta\boldsymbol{\theta}^{(k)} = J^{\dagger}(\boldsymbol{\theta}^{(k)}) \left(\mathbf{x}_d - f(\boldsymbol{\theta}^{(k)})\right)$$

with $J^{\dagger}$ being the pseudoinverse:

$$J^{\dagger} = J^{\top}(JJ^{\top})^{-1}$$

The update rule is as follows:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \Delta\boldsymbol{\theta}^{(k)}$$

and iteration continues until:

$$\|\mathbf{x}_d - f(\boldsymbol{\theta}^{(k)})\| < \epsilon$$

6.5.2 Sampling-Based Motion Planning using RRT*

Once a desired pose of the joint configuration $\boldsymbol{\theta}_d$ is found, a corresponding joint configuration trajectory is calculated by solving a constrained motion planning problem in the joint space. The underlying planner is RRT* (Rapidly Exploring Random Tree Star), a sampling-based algorithm that incrementally builds a tree of valid configurations and asymptotically converges to the optimal solution with respect to a given cost function, typically minimizing the path length or joint space distance.

Let $\boldsymbol{\theta}_{\text{start}} \in \mathbb{R}^n$ be the current joint configuration and $\boldsymbol{\theta}_{\text{goal}}$ be a valid configuration satisfying the inverse kinematics of the target pose. RRT* samples random configurations $\boldsymbol{\theta}_{\text{rand}}$ and constructs a tree $\mathcal{T}$ such that each edge represents a feasible transition under joint, velocity, and collision constraints. The planner selects the configuration $\boldsymbol{\theta}_{\text{nearest}} \in \mathcal{T}$ closest to $\boldsymbol{\theta}_{\text{rand}}$ and attempts to guide it with a new node $\boldsymbol{\theta}_{\text{new}}$.

At each iteration, RRT* rewires the tree to optimize the cumulative cost-to-come from the start to $\boldsymbol{\theta}_{\text{new}}$:

$$\mathcal{T} \leftarrow \text{Rewire}(\mathcal{T}, \boldsymbol{\theta}_{\text{new}})$$

As the number of samples increases, the RRT* solution converges to the shortest or cheapest feasible trajectory between the start and the goal configurations.

6.5.3 Constant Velocity Timing

To achieve a constant linear end-effector velocity, the Euclidean distance between consecutive waypoints is computed:

$$d_i = \|\mathbf{p}_i - \mathbf{p}_{i-1}\|$$

Given a desired velocity $v_d > 0$, the corresponding time step is as follows:

$$\Delta t_i = \frac{d_i}{v_d}, \quad t_i = t_{i-1} + \Delta t_i$$

This produces time-stamped joint configurations $\{(\boldsymbol{\theta}_i, t_i)\}$ ensuring approximately uniform Cartesian speed along the welding path.

6.5.4 Angle Continuity and Unwrapping

To avoid discontinuities in joint space motion due to angular wrapping (e.g. $0 \leftrightarrow 2\pi$), each joint angle trajectory is post-processed. Let $\boldsymbol{\theta}_{i-1}$ and $\boldsymbol{\theta}_i$ be two consecutive joint vectors. The relative difference is

$$\Delta\boldsymbol{\phi}_i = \boldsymbol{\theta}_i - \boldsymbol{\theta}_{i-1}$$

We unwrap each joint angle to ensure the following:

$$\Delta\phi_{i,j} \in (-\pi, \pi], \quad \forall j = 1, \ldots, n$$

The corrected angle is

$$\boldsymbol{\theta}_i^{\text{adj}} = \boldsymbol{\theta}_{i-1} + \text{unwrap}(\Delta\boldsymbol{\phi}_i)$$

This guarantees a smooth and continuous evolution of the joint values throughout the trajectory.

6.5.5 Resulting Trajectory

The final output is a time-parameterized joint-space trajectory:

$$\boldsymbol{\theta}(t), \quad t \in [t_0, t_N]$$

which satisfies: - Constant Cartesian velocity, - Smooth joint transitions, - Accurate inverse kinematics matching, - Feasibility under joint and workspace constraints. This trajectory is then passed on to the low-level controller for execution of the desired welding motion.

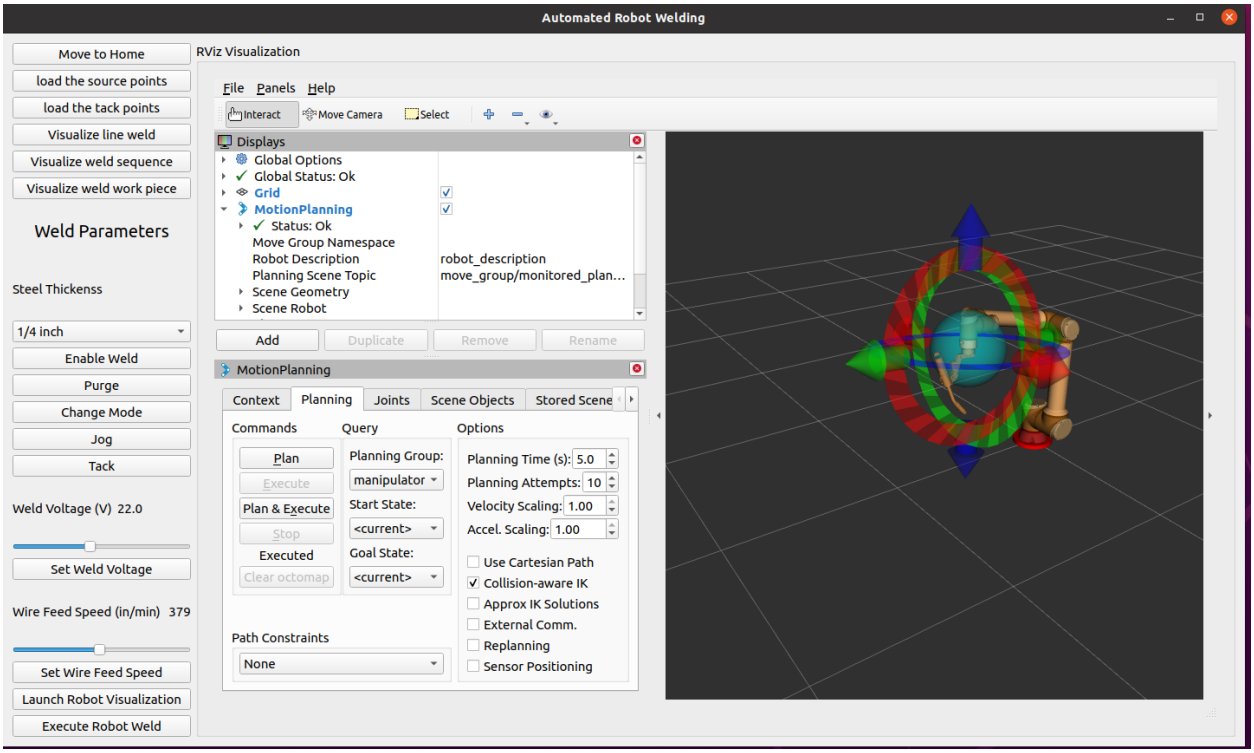

Figure 8: GUI Module for Automation.

7 Robotic Welding GUI

The operation of the entire cobotic welding sequence is performed using a custom Graphical User Interface (GUI) developed in Python with ROS (Robot Operating System) integration (see Fig. 8). The GUI acts as a central hub for interfacing with the welding robot, configuring weld parameters, visualizing the robot state in RViz, and executing planned welding trajectories.

The GUI is implemented using the `PyQt5` framework and interfaces with two main back-end modules. The `RobotRegistration` module is responsible for handling workspace registration, tack point loading, and mesh visualization. The `Execute_controller` module manages motion planning and trajectory execution. In addition, the GUI integrates RViz, a 3D visualization tool in ROS that allows for real-time display of the robot's configuration, environment, and sensor feedback.

Once the user completes the registration and visualization steps, the robot calls the execution logic from the `Execute Robot Weld` module. This module computes a Cartesian trajectory using the registered tack points and their associated orientations. The trajectory is generated to ensure smooth motion with constant linear velocity and the correct orientation of the end effector throughout the weld path. The corresponding joint trajectory is then sent to the robot controller for execution. The GUI provides a modular and user-friendly interface for controlling the complete robotic welding pipeline from calibration to final execution, reducing human error, improving weld repeatability, and supporting flexible operation within the workspace. The key elements of the GUI and their functionalities are provided in Table 1.

Section	Functionalities
Registration	`Move to Home` – Sends the robot to its home position. `Load Source Points` – Loads mesh reference points. `Load Tack Points` – Loads welding tack points.
Path Planning	`Line Weld` – Shows linear weld path in RViz. `Weld Sequence` – Displays numbered weld sequence. `Workpiece View` – Visualizes complete workpiece mesh.
Weld Parameters	`Jog` – Sends jog command to MIG controller. `Tack` – Triggers short weld pulse. Sliders for: **Voltage (10–38V)** and **Feed Rate (0–780 in/min)**. `Set Voltage` / `Set Feed Rate` – Apply settings.
Visualization & Execution	`RViz Viewer` – Shows robot state in real time. `Execute Weld` – Runs trajectory on robot.

Table 1: GUI Layout and Key Functionalities

References

[1] D. Curiel, F. Veiga, A. Suarez, and P. Villanueva, "Advances in robotic welding for metallic materials: Application of inspection, modeling, monitoring and automation techniques," *Metals*, vol. 13, no. 4, 2023.

[2] S. I. Wahidi, S. Oterkus, and E. Oterkus, "Robotic welding techniques in marine structures and production processes: A systematic literature review," *Marine Structures*, vol. 95, p. 103608, May 2024.

[3] M. Biesuz, T. Saunders, D. Ke, M. J. Reece, C. Hu, and S. Grasso, "A review of electromagnetic processing of materials (epm): Heating, sintering, joining and forming," *Journal of Materials Science & Technology*, vol. 69, pp. 239–272, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1005030220307106

[4] R. Goldiez. (2021) Welding robots: Types, advantages, and limitations. Accessed:2023. [Online]. Available: https://blog.hirebotics.com/guide-to-welding-robots

[5] (2022) 5 causes of contact tip burnback & how to solve them. Accessed:2023. [Online]. Available: https://americantorchtip.com/blog/5-causes-of-contact-tip-burnback/

[6] (2020) How to adjust mig welder wire feed roller tension,. Accessed:2023. [Online]. Available: https://americantorchtip.com/blog/5-causes-of-contact-tip-burnback/

[7] S. Woods. (2020) How to adjust mig welder wire feed roller tension,. Accessed:2023. [Online]. Available: https://fsmdirect.com/based-on-a-tip/

[8] Porosity in welding - defects / imperfections in welds. The Welding Institute. Accessed:2023. [Online]. Available: https://www.twi-global.com/technical-knowledge/job-knowledge/defects-imperfections-in-welds-porosity-042.aspx

[9] (2019, October) What is penetration in welding? - weldingtech.net. Welding Tech. Accessed: February 15, 2023. [Online]. Available: https://weldingtech.net/penetration/

[10] I. Noreen, A. Khan, and Z. Habib, "Optimal path planning using rrt* based approaches: A survey and future directions," *International Journal of Advanced Computer Science and Applications*, vol. 7, no. 11, 2016. [Online]. Available: http://thesai.org/Publications/ViewPaper?Volume=7&Issue=11&Code=ijacsa&SerialNo=14

[11] F. Ducho, A. Babinec, M. Kajan *et al.*, "Path planning with modified a star algorithm for a mobile robot," *Procedia Engineering, Modelling of Mechanical and Mechatronic Systems*, vol. 96, pp. 59–69, Jan. 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S187770581403149X

[12] K. M. Lynch and F. C. Park, *Modern Robotics: Mechanics, Planning, and Control.* Cambridge, UK: Cambridge University Press, 2017.

[13] Truly autonomous robots powered by ai technology. Path Robotics. [Online]. Available: https://www.path-robotics.com/